AIR QUALITY ANALYSIS

India • 2015 – 2020

A Data-Driven Look at Pollution, Seasons & the COVID-19 Lockdown Effect

Prepared by Fathima Suzain | Python • Pandas • Matplotlib • Seaborn

29,531 → 24,850 Rows cleaned	**26** Cities analyzed	**6 yrs** 2015–2020 span	**5** Key insights found

Source: Kaggle — Air Quality Data in India, originally published by the Central Pollution Control Board (CPCB), Government of India.

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■ 1. Project Overview

This project analyzes daily air quality readings across 26 major Indian cities from 2015 to 2020. Rather than a plain dataset dump, the goal was a genuine analytical story: which cities are most polluted, how pollution shifted over time, whether India's 2020 COVID-19 lockdown measurably improved air quality, and which pollutants drive overall air quality most strongly.

■ **Source**	Kaggle — rohanrao/air-quality-data-in-india
■ **Origin**	Central Pollution Control Board (CPCB), Govt. of India
■ **Raw size**	29,531 rows × 16 columns
■ **Cleaned size**	24,850 rows × 15 columns — zero missing values
■ **Cities**	26, including Delhi, Mumbai, Ahmedabad, Chennai, Kolkata, Bengaluru
■ **Time span**	2015 – 2020 (2020 data available through July)
■ **Tools**	Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

■ 2. Data Cleaning Process

Real-world sensor data is inherently messy — stations go offline, equipment breaks, and some pollutants weren't tracked in earlier years. Each cleaning decision below was made deliberately, not just to "fill blanks."

✂ Dropped Xylene column

61% of values were missing (18,109 of 29,531 rows) — too unreliable to impute, so the column was removed entirely.

■ City-wise median imputation

PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene were filled using each city's own median, since baseline pollution varies hugely by city.

■ Overall median fallback

A few columns still had gaps where a whole city had zero readings for that pollutant — filled with the overall dataset median instead.

■ AQI / AQI_Bucket handling

AQI is a calculated field, not a raw reading, so it can't be median-filled. Rows missing AQI (4,681, ~16%) were dropped instead.

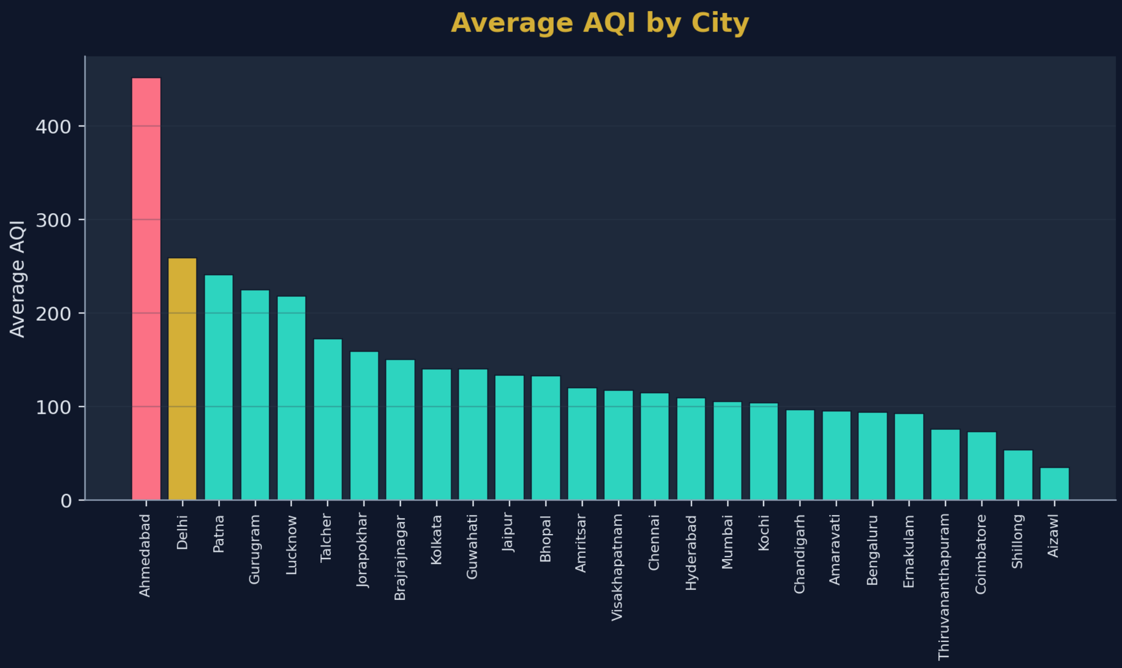
■ Feature engineering

Date converted to proper datetime; Year and Month extracted for trend and seasonal analysis.

BEFORE		AFTER
29,531 rows 16 columns Missing values in 10+ columns		24,850 rows 15 columns **Zero missing values ✓**

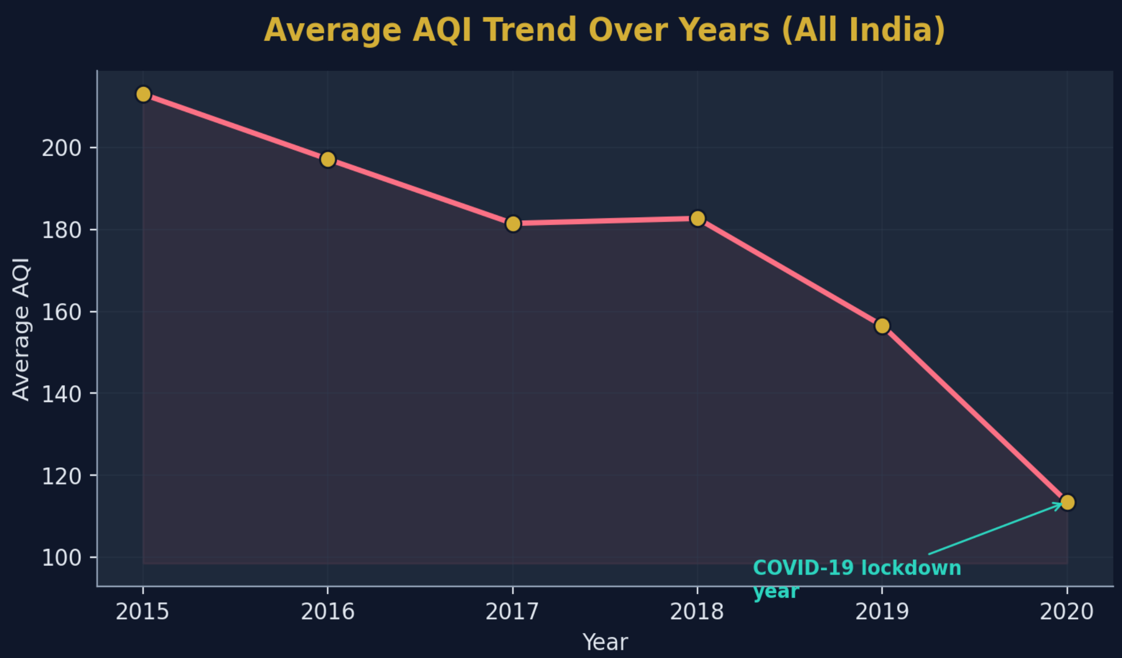
3. Key Insights

Insight 1 — Most Polluted Cities



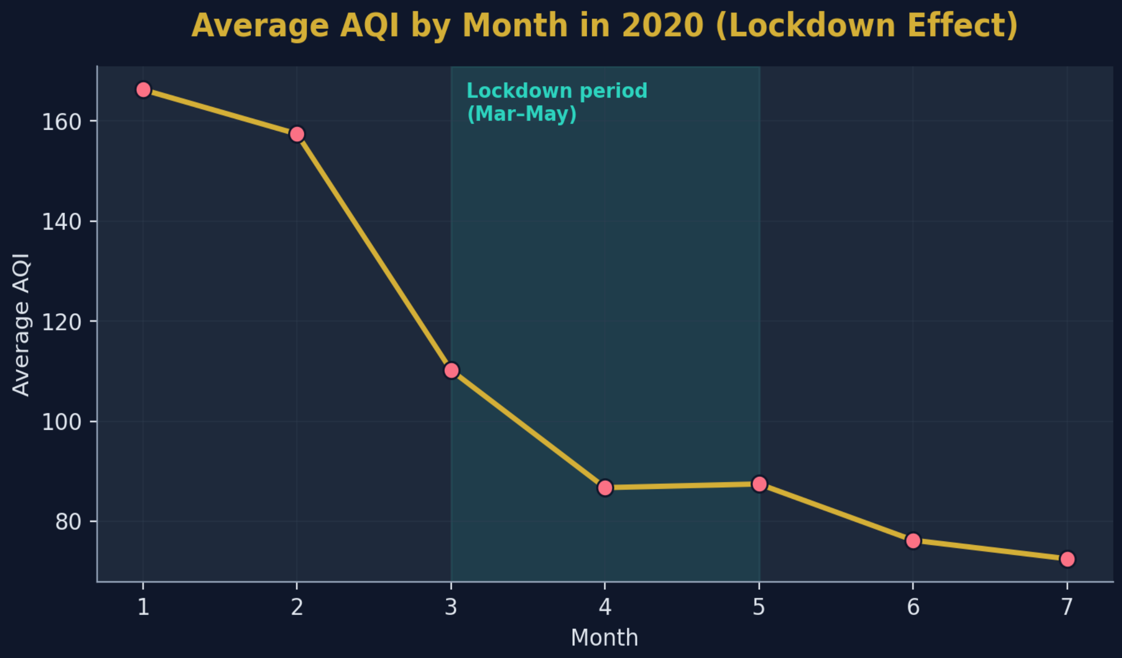
Ahmedabad recorded the highest average AQI (452), notably ahead of **Delhi** (259), which is more commonly cited as India's most polluted city in the media. Both cities have 6 full years of data, so this isn't a sample-size artifact — it's a genuine finding.

Insight 2 — National AQI Trend, 2015–2020



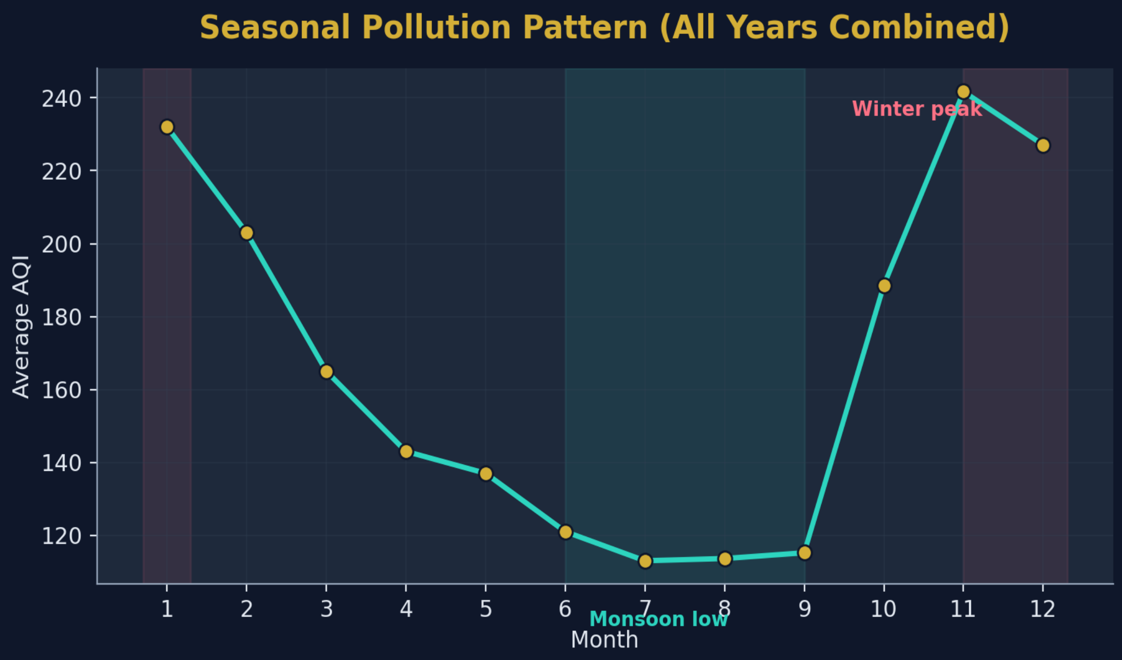
Average AQI declined steadily nationwide, from ~213 (2015) to ~113 (2020) — with a notably steeper drop between 2019 and 2020 compared to the gradual decline in earlier years.

■ Insight 3 — The COVID-19 Lockdown Effect



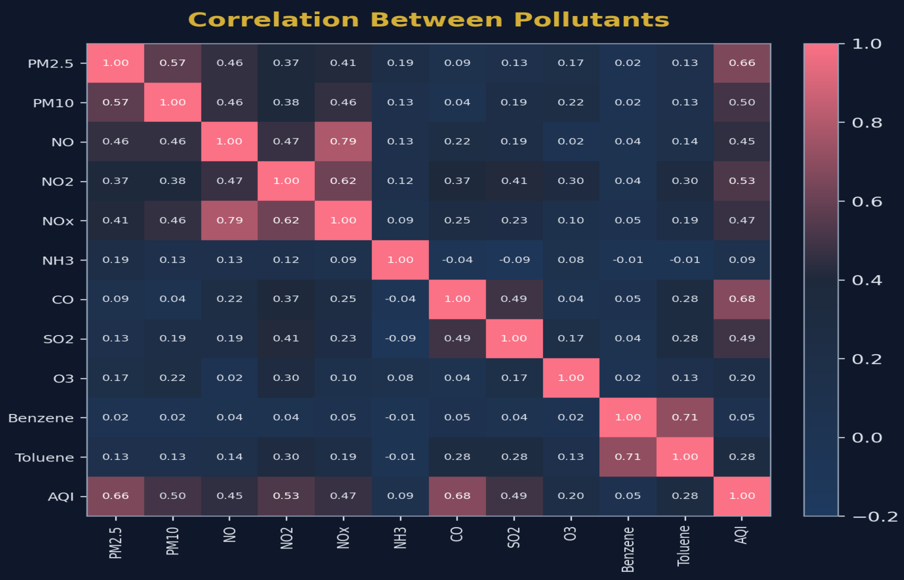
Breaking 2020 down by month confirms it: AQI fell from ~166 in Jan/Feb to just ~87 in April — the strictest lockdown month — before gradually recovering. The timing lines up precisely with reduced vehicular and industrial activity.

❄ Insight 4 — Seasonal Pollution Pattern



Averaged across all six years, AQI traces a clear U-shape: peaking Nov–Jan due to winter smog and crop stubble burning, and dropping to its lowest in the monsoon months of Jun–Sep, when rain and wind naturally disperse pollutants.

■ Insight 5 — Pollutant Correlations



PM2.5 (0.66) and **CO** (0.68) show the strongest correlation with overall AQI, confirming fine particulate matter and combustion emissions as the primary pollution drivers. NO/NOx (0.79) and Benzene/Toluene (0.71) also move tightly together, as expected from shared vehicle-exhaust origins.

■ 4. Interview Walkthrough Summary

A quick recap of the project flow, for explaining it end-to-end in an interview:

1. Loaded city_day.csv (29,531 rows) with Pandas; inspected structure via .info() and .isnull().sum()
2. Cleaned the dataset: dropped Xylene, imputed pollutants city-wise then by overall median, dropped rows missing AQI → 24,850 clean rows, zero missing values
3. Engineered Year and Month columns from Date for time-based analysis
4. Analyzed most polluted cities by average AQI, validated against data availability per city
5. Plotted the national AQI trend 2015–2020 and investigated the sharp 2020 drop
6. Confirmed the COVID-19 lockdown effect by breaking 2020 down month-by-month
7. Identified India's seasonal pollution pattern (winter peak, monsoon low) across all years
8. Built a pollutant correlation heatmap to identify the strongest drivers of AQI

■ 5. Author & Links

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✦ Documentation generated as part of a personal data analytics portfolio project ✦